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CS 598 Foundations of Data Curation

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# Progress Report: Predicting F1 Race Winners through Data Curation

**Initial Proposal Feedback Considerations**

With our project proposal graded and the following feedback provided:

> Overall a well-written proposal with an interesting research topic. Given that you will be working as a group, I think the scale of the project is doable. Looking at the proposal and the data sources, a problem you guys might run into is - the project being larger than expected (too many data sources and columns to integrate and take into consideration). It's okay to make a stricter assumption or eliminate some variable such as tire type in order to make it more manageable. Feel free to make a campuswire post when you run into any issues and discuss with TAs.

We agree with the TA that there may be too many data sources to integrate. Given that, we are still hoping to integrate the Pirelli data into the pipeline, however, if not possible, we are willing to forgo that source and drop tyre compounds from consideration entirely.

**Status of Deliverables**

A GitHub repository was created with structured folders **(Week 1a & b).** Next, we tested API access and generated raw data using the endpoints **(Week 2a & c)**. This was done to get a feel for the data to help us develop a proper schema. We also manually created the table of the data provided in the Pirelli newsroom article; this was done just in case we want to start working with the data **(Week 2b)**. Given that all team members have a lot of experience with SQL, we decided to create a schema using SQL **(Week 3b)**. The schema is still in the works but we are confident with the overall structure; there may just be certain features added or removed as we explore further **(Week 3a)**. Finally, we started working on the data cleaning aspect of the project which still needs more work as we change the schema **(Week 4)**, but so far we have about 75% of the data cleaned up; it would be more if not for the challenges listed below.

**Challenges**

<u>OpenF1 Access:</u> The reliability of the OpenF1 API, a key component of our data pipeline, has been somewhat unreliable, which is a significant concern we need to address going forward. Specifically, the owner for OpenF1 API mentioned the possibility of turning off public access due to increasing cost maintenance. We were unable to access the API earlier; however, as of (10/26/2025), we are now able to access it again. This intermittent access highlights the ongoing risk of relying on a public-facing service with potential financial pressures. We will continue to monitor the OpenF1 API's status and develop plans to mitigate this risk.

Filled out this form: https://tally.so/r/w2yWDb. Worst case scenario, we'll pay $20 to use the API if the admin provides that option.

OpenWeather does not have Public API Access: For this reason, we switched over to OpenMeteo.

**Timeline Updates**

1. Further Data Cleaning **[Week of 10/27]**
    a. Clean the datasets: handle missing values, fix time forms, find primary keys, etc… - Each of us will split the datasets and handle one
2. Pirelli Data Automation (OPTIONAL) **[Week 10/27 or 11/3]**
    a. We still CAN make a script that gets the Pirelli data automatically. – Fatih
3. Document the Data **[Week of 11/3]**
    a. Document why we chose the schema in step 3. - As a group.
    b. Create a data dictionary where we explain the variables, their units and meanings. - Each person for the dataset we chose.
    c. Write clear documentation on how the data was transformed in step 4 - As a group.
4. Exploration & Predictive Analysis **[Week of 11/10]**
    a. Basic descriptive statistics - Fatih
    b. Generate plots for data? (not necessary for curation but if we want to explore the "usage of data" section. - Kunal
    c. Analyze by using statistical methods to offer an answer to the research question. - Kunal & Navtej
5. Automate the workflow **[Week of 11/17]**
    a. Write scripts to automate acquiring, cleaning, and integrating/modeling the data. - Fatih & Kunal
    b. Testing and debug workflow. - Fatih & Navtej
6. Preserve & Share - All of us **[Week of 11/24]**
    a. Archive datasets, scripts, and documentation in GitHub
    b. Make sure the archived information is clear and structured
    c. Make sure version control is up to date (this should be something done in every step)
    d. Package into a ZIP file?
7. Write Report **[Week of 12/1]**

**Deliverables**

General GitHub Repository: https://github.com/kunaldaftari1/cs_598_f1_project

Raw Exploration Data: https://github.com/kunaldaftari1/cs_598_f1_project/tree/main/data
- FastF1 Raw Sample Data
- OpenF1 Raw Sample Data
- OpenMeteo Raw Sample Data

SQL Schema: https://github.com/kunaldaftari1/cs_598_f1_project/deliverables/schema.sql

Pirelli Manual Data:
https://github.com/kunaldaftari1/cs_598_f1_project/blob/main/deliverables/pirelli_tire_data.py

Data Cleaning Scripts:
https://github.com/kunaldaftari1/cs_598_f1_project/blob/main/scripts/data_cleaning/all_races.ipynb

**New References (APA7)**

Open Meteo API Documentation: https://open-meteo.com/

*Free weather API*. Open. (n.d.-b). https://open-meteo.com/

**Old References (APA7)**

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Godefroy, B. (n.d.). *Introduction*. OpenF1 API | Real-time and historical Formula 1 data. https://openf1.org/

*One call API 3.0*. Product Concept. (n.d.). https://openweathermap.org/api/one-call-3

Pirelli. (2025a, February 6). *All the 2024 season stats: From the Earth to the Moon (almost) with formula 1's Pirelli Tyres*. Pirelli Newsroom. https://www.pirelli.com/global/en-ww/race/racingspot/formula-1/all-the-2024-f1-season-stats-155524/

Pirelli. (2025b, March 17). *Formula 1 Louis Vuitton Australian Grand Prix 2025*. Pirelli Newsroom. https://www.pirelli.com/global/en-ww/emotions-and-numbers/formula-1-louis-vuitton-australian-grand-prix-2025-159448/

Pirelli. (2025c, August 1). *Changes and status quo when it comes to compound choices for the rest of the season*. Pirelli Newsroom. https://press.pirelli.com/changes-and-status-quo-when-it-comes-to-compound-choices-for-the-rest-of-the-season0/

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Schaefer, P. (n.d.). *TheOehrly/Fast-F1: FastF1 is a python package for accessing and analyzing formula 1 results, schedules, timing data and telemetry*. GitHub. https://github.com/theOehrly/Fast-F1